2nd Midterm · Unit 5 · Fourier series and convergence · 5 points · 45 min · Standard exam level

P10 — Full Fourier series on (-pi, pi) of the function equal to 0 on the left half and x^2 on the right half: coefficients, Dirichlet values and two numerical series

Let f be defined on $(-\pi,\pi)$ by

$$f(x)=\begin{cases}0, & -\pi<x<0,\\ x^2, & 0\le x<\pi,\end{cases}$$

and extended to $\mathbb{R}$ as a 2π-periodic function.

(a) Describe the graph of the periodic extension on $[-3\pi,3\pi]$: where is it continuous, where does it jump and by how much? Is f piecewise smooth?

(b) Compute the Fourier series $\dfrac{a_0}{2}+\displaystyle\sum_{n\ge1}\left(a_n\cos nx+b_n\sin nx\right)$ of f.

(c) Using the Dirichlet theorem, give the sum of the series at $x=0$, $x=\pi$, $x=-\pi$, $x=\pi/2$ and $x=-\pi/2$.

(d) Deduce the sums of $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^2}$ and $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}$, and check that the two results are consistent with each other.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns: the coefficients a_0, a_n, b_n ($n\ge1$). Independent variable $x\in\mathbb{R}$, period 2π ($L=\pi$).
- f is bounded, and on each of $(-\pi,0)$ and $(0,\pi)$ it is a polynomial: f and f' are piecewise continuous, so f is piecewise smooth and the Dirichlet theorem applies.
- No symmetry: f is neither even nor odd, so both a_n and b_n are needed. Since $f=0$ on $(-\pi,0)$, every integral reduces to $(0,\pi)$.

Phase 2. Canonical formulation

Euler formulas on $(-\pi,\pi)$:

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos nx\,dx\ (n\ge0),\qquad b_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\sin nx\,dx\ (n\ge1).$$

Dirichlet theorem: for a piecewise smooth 2π-periodic f, the series converges at every x to $\frac{1}{2}\big[f(x^+) + f(x^-)\big]$.

Part (a) · 1 point

Periodic extension, continuity and jumps

Step 1. One period

On $(-\pi, 0)$ the graph is the segment $y = 0$; on $[0, \pi)$ it is the arc of $y = x^2$, rising from $(0, 0)$ to height π^2 as $x \to \pi^-$.

Step 2. Continuity at the origin

$f(0^-) = 0$, $f(0^+) = 0^2 = 0$: continuous. The one-sided derivatives are 0 (left) and $2 \cdot 0 = 0$ (right): the graph even has no corner at $x = 0$; only f'' jumps there (from 0 to 2).

Step 3. Jumps at the odd multiples of pi

The extension satisfies $f(x + 2\pi) = f(x)$. At $x = \pi$: $f(\pi^-) = \lim_{x \to \pi^-} x^2 = \pi^2$, while $f(\pi^+) = f(-\pi^+) = 0$. So the extension jumps down by π^2 at $x = \pm\pi, \pm 3\pi$. The graph on $[-3\pi, 3\pi]$ is three copies of the period: flat on $(-3\pi, -2\pi)$, arc on $(-2\pi, -\pi)$, flat on $(-\pi, 0)$, arc on $(0, \pi)$, flat on $(\pi, 2\pi)$, arc on $(2\pi, 3\pi)$, with a vertical drop from π^2 to 0 at each odd multiple of π.

KEY RESULTS

- On each period: zero on $(-\pi, 0)$, the parabola arc x^2 from 0 up to π^2 on $(0, \pi)$.
- Continuous at $x = 2k\pi$ ($f(0^-) = f(0^+) = 0$, and even $f'(0^\pm) = 0$).
- Jumps at $x = (2k+1)\pi$: $f(\pi^-) = \pi^2$, $f(\pi^+) = 0$, jump of size $-\pi^2$.
- Piecewise smooth: yes.

Part (b) · 2 points

Fourier coefficients

Step 1. Mean value

$$a_0 = \frac{1}{\pi}\int_0^{\pi} x^2\,dx = \frac{1}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi} = \frac{1}{\pi}\cdot\frac{\pi^3}{3} = \frac{\pi^2}{3}.$$

Step 2. Two auxiliary integrals ($n \geq 1$)

By parts with $u = x$, $dv = \sin nx\,dx$, $du = dx$, $v = -\frac{\cos nx}{n}$:

$$\int x \sin nx\,dx = -\frac{x\cos nx}{n} + \frac{1}{n}\int \cos nx\,dx = -\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}.$$

By parts with $u = x$, $dv = \cos nx\,dx$, $du = dx$, $v = \frac{\sin nx}{n}$:

$$\int x \cos nx\,dx = \frac{x\sin nx}{n} - \frac{1}{n}\int \sin nx\,dx = \frac{x\sin nx}{n} + \frac{\cos nx}{n^2}.$$

Step 3. Cosine coefficients

By parts with $u = x^2$, $dv = \cos nx\,dx$, $du = 2x\,dx$, $v = \frac{\sin nx}{n}$, and then Step 2:

$$\int x^2 \cos nx\,dx = \frac{x^2\sin nx}{n} - \frac{2}{n}\int x\sin nx\,dx = \frac{x^2\sin nx}{n} + \frac{2x\cos nx}{n^2} - \frac{2\sin nx}{n^3}.$$

Barrow's rule on $[0,\pi]$, with $\sin n\pi = \sin 0 = 0$ and $\cos n\pi = (-1)^n$:

$$\int_0^{\pi} x^2\cos nx\,dx = \left[0 + \frac{2\pi(-1)^n}{n^2} - 0\right] - \left[0 + 0 - 0\right] = \frac{2\pi(-1)^n}{n^2}, \qquad a_n = \frac{2(-1)^n}{n^2}.$$

Step 4. Sine coefficients

By parts with $u = x^2$, $dv = \sin nx\,dx$, $du = 2x\,dx$, $v = -\frac{\cos nx}{n}$, and then Step 2:

$$\int x^2 \sin nx\,dx = -\frac{x^2\cos nx}{n} + \frac{2}{n}\int x\cos nx\,dx = -\frac{x^2\cos nx}{n} + \frac{2x\sin nx}{n^2} + \frac{2\cos nx}{n^3}.$$

Barrow's rule on $[0,\pi]$:

$$\int_0^{\pi} x^2 \sin nx\, dx = \Big[-\frac{\pi^2(-1)^n}{n} + 0 + \frac{2(-1)^n}{n^3} \Big] - \Big[0 + 0 + \frac{2}{n^3} \Big] = -\frac{\pi^2(-1)^n}{n} + \frac{2\big((-1)^n - 1\big)}{n^3}.$$

Dividing by π:

$$b_n = \frac{(-1)^{n+1}\pi}{n} + \frac{2\big((-1)^n - 1\big)}{\pi n^3} = \begin{cases} \dfrac{\pi}{n} - \dfrac{4}{\pi n^3}, & n \text{ odd}, \\ -\dfrac{\pi}{n}, & n \text{ even}. \end{cases}$$

Step 5. The series

$$f(x) \sim \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \left[\frac{2(-1)^n}{n^2} \cos nx + \left(\frac{(-1)^{n+1}\pi}{n} + \frac{2\big((-1)^n - 1\big)}{\pi n^3} \right) \sin nx \right].$$

First terms: $\frac{\pi^2}{6} - 2\cos x + \left(\pi - \frac{4}{\pi}\right)\sin x + \frac{1}{2}\cos 2x - \frac{\pi}{2}\sin 2x - \frac{2}{9}\cos 3x + \dots$

KEY RESULTS

- $a_0 = \dfrac{\pi^2}{3}$ (so the constant term is $\dfrac{\pi^2}{6}$).
- $a_n = \dfrac{2(-1)^n}{n^2}$: $a_1 = -2$, $a_2 = \frac{1}{2}$, $a_3 = -\frac{2}{9}$.
- $b_n = \dfrac{(-1)^{n+1}\pi}{n} + \dfrac{2\big((-1)^n - 1\big)}{\pi n^3}$: $b_n = \dfrac{\pi}{n} - \dfrac{4}{\pi n^3}$ for n odd, $b_n = -\dfrac{\pi}{n}$ for n even.

Part (c) · 1 point

Values of the series by the Dirichlet theorem

Step 1. Points of continuity

$x = 0$: f is continuous there, sum $f(0) = 0$. $x = \pi/2$: inside $(0, \pi)$, sum $(\pi/2)^2 = \pi^2/4$. $x = -\pi/2$: inside $(-\pi, 0)$, sum 0.

Step 2. The endpoints are jump points

For the periodic extension, $x = \pi$ and $x = -\pi$ are the same kind of point: $f(\pi^-) = \pi^2$, $f(\pi^+) = f(-\pi^+) = 0$. The series converges to

$$\frac{f(\pi^+)+f(\pi^-)}{2}=\frac{0+\pi^2}{2}=\frac{\pi^2}{2},$$

the same value at $x=-\pi$ (each term is 2π-periodic, so the series takes equal values at $\pm\pi$). It is not $f(\pi)$ of either formula: the series does not "choose" a side.

KEY RESULTS

- $x=0$: 0. $x=\pm\pi$: $\dfrac{\pi^2}{2}$. $x=\pi/2$: $\dfrac{\pi^2}{4}$. $x=-\pi/2$: 0.

Part (d) · 1 point

Two numerical series

Step 1. Evaluation at x = 0

$\cos 0=1$, $\sin 0=0$, and by (c) the sum is 0:

$$0=\frac{\pi^2}{6}+\sum_{n=1}^{\infty}\frac{2(-1)^n}{n^2}\implies\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}=-\frac{\pi^2}{12}\implies\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}=1-\frac{1}{4}+\frac{1}{9}-\cdots=\frac{\pi^2}{12}.$$

Step 2. Evaluation at x = pi

$\cos n\pi=(-1)^n$, $\sin n\pi=0$, and by (c) the sum is $\pi^2/2$:

$$\frac{\pi^2}{2}=\frac{\pi^2}{6}+\sum_{n=1}^{\infty}\frac{2(-1)^n(-1)^n}{n^2}=\frac{\pi^2}{6}+2\sum_{n=1}^{\infty}\frac{1}{n^2}\implies\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{1}{2}\left(\frac{\pi^2}{2}-\frac{\pi^2}{6}\right)=\frac{\pi^2}{6}.$$

Step 3. Consistency check

Subtracting the two series, the odd terms cancel and the even ones double: $\sum \frac{1}{n^2} - \sum \frac{(-1)^{n+1}}{n^2} = 2\sum_{m\geq 1} \frac{1}{(2m)^2} = \frac{1}{2}\sum_{m\geq 1} \frac{1}{m^2}$. With the values found: $\frac{\pi^2}{6} - \frac{\pi^2}{12} = \frac{\pi^2}{12}$ and $\frac{1}{2}\cdot\frac{\pi^2}{6} = \frac{\pi^2}{12}$. They agree.

KEY RESULTS

- $x = 0$: $0 = \dfrac{\pi^2}{6} + 2\sum \dfrac{(-1)^n}{n^2}$, so $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$.
- $x = \pi$: $\dfrac{\pi^2}{2} = \dfrac{\pi^2}{6} + 2\sum \dfrac{1}{n^2}$, so $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.
- Consistency: $\dfrac{\pi^2}{6} - \dfrac{\pi^2}{12} = 2\displaystyle\sum_{n \text{ even}} \frac{1}{n^2} = \frac{1}{2}\cdot\frac{\pi^2}{6}$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $a_n = O(1/n^2)$ because f is continuous at 0, while $b_n \sim (-1)^{n+1}\pi/n$ decays only like $1/n$: that slow part is produced by the jump of size π^2 at $x = \pm\pi$ (a jump of size J at $x = \pi$ always gives a contribution $\frac{J}{\pi}\frac{(-1)^n}{n}$ with $J = -\pi^2$ here). Near the jump the partial sums overshoot by about 9% of the jump (Gibbs phenomenon, **extension**: the overshoot of partial sums near a jump, which does not vanish as $N \to \infty$).
- **Checks:** the antiderivatives of Steps 3 and 4 give back $x^2\cos nx$ and $x^2\sin nx$ when differentiated; partial sums with 4000 terms reproduce 0 at $x = 0$, $\pi^2/2$ at $x = \pi$ and $\pi^2/4$ at $x = \pi/2$ to 10^{-3}; the mean value $a_0/2 = \pi^2/6$ equals $\frac{1}{2\pi}\int_{-\pi}^{\pi} f\,dx$.
- **Common mistakes in the exam:** (1) dividing by 2π instead of π in a_n, or forgetting that the constant term is $a_0/2$; (2) evaluating the series at $x = \pi$ as $f(\pi^-) = \pi^2$ instead of the midpoint $\pi^2/2$, which gives $\sum 1/n^2 = 5\pi^2/12$ (wrong); (3) dropping the term $2((-1)^n - 1)/(\pi n^3)$ of b_n, which only lives on odd n; (4) treating f as even or odd and discarding one family of coefficients.